

Problem 2.16

For non-negative integer n , define

$$i_n = \frac{A(n) - A(n-1)}{A(n-1)}.$$

Show that

$$(1 + i_n)^{-1} = \frac{A(n-1)}{A(n)}.$$

Solution

Starting from the definition,

$$i_n = \frac{A(n) - A(n-1)}{A(n-1)}.$$

Multiply both sides by $A(n-1)$:

$$i_n A(n-1) = A(n) - A(n-1).$$

Now add $A(n-1)$ to both sides:

$$i_n A(n-1) + A(n-1) = A(n).$$

Factor $A(n-1)$ on the left-hand side:

$$A(n-1)(i_n + 1) = A(n).$$

Divide both sides by $A(n)$:

$$\frac{A(n-1)}{A(n)} = \frac{1}{1 + i_n}.$$

Therefore,

$$\boxed{\frac{A(n-1)}{A(n)} = (1 + i_n)^{-1}}.$$